

Arnav Parekar

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SUMMARY

CS student with strong foundation in software engineering principles and hands-on experience developing production-grade applications. Proven ability to collaborate with cross-functional teams, contribute to code reviews, and apply problem-solving skills across multiple programming languages including Java, Python, and C++. Eager to learn new technologies and contribute to real-world projects.

EDUCATION

Vellore Institute of Technology, Chennai
Bachelor of Technology in Computer Science & Engineering

CGPA : 9.19/10
July 2023 – May 2027

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, Dart
Software Engineering: Software Development Lifecycle, Code Reviews, Git/GitHub, Agile Methodologies
Backend Development: Flask, FastAPI, REST APIs, Microservices Architecture
Full-Stack: React.js, Vue.js, Flutter, Next.js, Cross-Platform Development
Cloud & Deployment: Google Cloud Platform, Microsoft Azure, Docker, CI/CD
Additional: SQL, Machine Learning, Network Security, Linux/Unix

EXPERIENCE

- OT Cybersecurity Intern** | *OSICS Technologies, Pune* June 2025 – July 2025
- Applied cybersecurity standards to **reduce potential vulnerabilities** in ICS network simulations
 - Gained hands-on experience with OT network architectures and ICS protocols (**Modbus, DNP3, OPC UA**) through the **LabShock** tool, analyzing traffic and **simulating attack scenarios** in a containerized SCADA setup
- Mathematics Club Technical Lead** | *VIT Chennai* July 2024 – June 2025
- Leading** technical initiatives, organizing workshops, and **mentoring** students in advanced mathematical problem-solving thus reaching out to more than **300 students**

PROJECTS

- TravelShield** | *React, Azure, Flask, Flutter, Firebase* [Link](#) | Nov 2024 – Feb 2025
- Developed travel health platform with **dual interfaces** - React web dashboard for airport authorities and Flutter mobile app for global travelers, demonstrating cross-platform UI expertise and scalable architecture
 - Designed scalable Flutter mobile application with multilingual support (5+ languages) and Firebase integration, handling **50+ concurrent** health assessments with **under 3s** response time
- InsightDoc** | *FAISS, RAG, Fast API* [Link](#) | Aug 2025 – Sep 2025
- Built comprehensive test suite for an intelligent document processing system, validating **FastAPI endpoints** handling **20+ concurrent enterprise queries** with sub-**5 second** response requirements
 - Implemented automated testing for the Retrieval-Augmented Generation pipeline, ensuring **90%+ accuracy** in document processing workflows through systematic test case design and validation
- FloatChat** | *RAG, Knowledge-based Graphs, Explainable AI* [Link](#) | Aug 2025 – Dec 2025
- Developed end-to-end RAG-powered chatbot integrating **MiniLM embeddings, FAISS semantic search**, and **knowledge graphs** for domain-specific Q&A with explainable AI capabilities documenting retrieval sources and confidence levels
 - Engineered **three-layer reasoning pipeline** combining RAG retrieval, row-column alignment, and LLM inference, building scalable data pipelines with ETL processes for **NetCDF data transformation** and Supabase storage
 - Orchestrated **MCP-based agentic AI** system for automated workflows with backend API services, implementing comprehensive testing protocols and monitoring mechanisms for production deployment

PATENTS

System for Monitoring Vehicle Speed in Traffic Environments and Method of Speed Estimation | *Patent Filed & Published* | *Application No: 202541130222*